

Removal of developmentally regulated microexons has a minimal impact on larval zebrafish brain morphology and function

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This **important** work examines how microexons contribute to brain activity, structure, and behavior. The authors find that loss of microexon sequences generally has subtle impacts on these metrics in larval zebrafish, with few exceptions. The evidence is **solid**, using modern high-throughput phenotyping methodology in zebrafish. Overall, this work will be of interest to neuroscientists and generate further studies of interest to the field.

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Abstract

Microexon splicing is a vertebrate-conserved process through which small, often in-frame, exons are differentially included during brain development and across neuron types. Although the protein sequences encoded by these exons are highly conserved and can mediate interactions, the neurobiological functions of only a small number have been characterized. To establish a more generalized understanding of their roles in brain development, we used CRISPR/Cas9 to remove 45 microexons in zebrafish and assessed larval brain activity, morphology, and behavior. Most mutants had minimal or no phenotypes at this developmental stage. Among previously studied microexons, we uncovered baseline and stimulus-driven phenotypes for two microexons (meA and meB) in *ptprd* and reduced activity in the telencephalon in the *tenm3* B₀ isoform. Although mild neural phenotypes were discovered for several microexons that have not been previously characterized, including in *ppp6r3*, *sptan1*, *dop1a*, *rapgef2*, *dctn4*, *vti1a*, and *meaf6*. This study establishes a general approach for investigating conserved alternative splicing events and prioritizes microexons for downstream analysis.

Introduction

Alternative splicing greatly expands proteome diversity and can impact brain development and disease (Vuong et al., 2016). As most metazoan proteins are represented by multiple isoforms, defining the functional significance of individual splicing events is a vast challenge. Mouse studies do not have the throughput to assess the neural role of 10s or 100s of isoforms, and cell culture does not recapitulate the complex development of an animal brain.

One class of particularly conserved and developmentally regulated spliced exons is microexons or mini-exons. These small exons, defined here as 3-30 nucleotides or 1-10 amino acids in length, are either gained or lost in transcripts in the brain compared to other tissues. While most are in-frame, this pathway can also trigger nonsense-mediated decay to influence protein levels (Lin et al., 2020). Although these neuron-specific exons were recognized almost 40 years ago (Brugge et al., 1985; Martinez et al., 1987), their discovery and annotation have continued into recent years (Parada et al., 2021). Thus far, hundreds have been identified. Although more than one splicing regulator has been implicated in the alternative inclusion of microexons in neurons (Ciampi et al., 2022; Y. I. Li et al., 2015), many are controlled by the vertebrate-specific protein SRRM4 (Irimia et al., 2014; Quesnel-Valli  res et al., 2015). Srrm4 interacts with Srsf11 and Rnps1 at polypyrimidine sequences upstream of microexons to regulate their inclusion (Gonatopoulos-Pournatzis et al., 2018).

Proteins that regulate and contain microexons are involved in neurodevelopmental disorders. In the brains of individuals with autism spectrum disorder (ASD), the level of SRRM4 is reduced, leading to the misregulated splicing of multiple microexons (Irimia et al., 2014). Mice haploinsufficient for *Srrm4* display reduced social interaction, sensory hypersensitivity, and impaired synaptic transmission (Quesnel-Valli  res et al., 2015). Furthermore, *Srrm4* is involved in regulating the ASD-relevant KCC2-dependent GABAergic excitatory to inhibitory switch (Nakano et al., 2019), and a mouse model of ASD with disrupted *Pten* has decreased *Srrm4* expression and microexon inclusion (Thacker et al., 2020). Loss-of-function variants in its interacting partner SRSF11 have been implicated in ASD in humans (C Yuen et al., 2017). While the behavioral roles of only a few microexons have been studied in detail (Gonatopoulos-Pournatzis and Blencowe, 2020), the elimination of one from *Eif4g1* altered synaptic plasticity and social behavior in mice (Gonatopoulos-Pournatzis et al., 2020). Many genes containing microexons also lead to neurodevelopmental disorders when mutated in humans. Examples include *CASK*-mediated microcephaly and cerebellar hypoplasia, *SPTAN1*-mediated childhood-onset epilepsy, and *MAPK8IP3*-mediated intellectual disability (Giacomini et al., 2021; Platzer et al., 2019; Syrbe et al., 2017). Taken together, this splicing program, the genes with microexons, and microexons themselves are important to enable proper neural development.

Although most microexons are unstudied, detailed structural and functional characterization of a small subset has revealed involvement in synapse development. The most well-studied are found in the receptor-type tyrosine phosphatase (e.g., *ptprf*) and teneurin (e.g., *tenm4*) gene families. Recognized in the 1990s (O'Grady et al., 1994; Pulido et al., 1995), two microexons in presynaptic PTP   (*Ptp  *), PTP   (*Ptp  *), and LAR (*Ptp  *) modulate *trans*-synaptic interfaces and determine selective interaction with postsynaptic partners (Lin et al., 2018; Y. Li et al., 2015; Park et al., 2020; Um et al., 2014; Yamagata et al., 2015; Yim et al., 2013). Similarly, differential inclusion of two conserved microexons in teneurins can modulate *trans*-synaptic homophilic or heterophilic interactions through splicing-dependent, large conformational changes (Berns et al., 2018; Gogou et al., 2024; Li et al., 2020). While studying these cell adhesion molecules has highlighted how microexons can trigger differential protein-protein interactions, little is known about the developmental consequences of these biochemical switches in other protein classes.

Identifying the neurobiological roles of many individual microexons is a challenge. Studies of splicing regulators such as *srrm4* impact the entire splicing program, making it impossible to determine the importance of individual microexons to protein function. Furthermore, microexons could still be differentially included in a *srrm4* regulatory mutant via compensation by other splicing factors, such as its paralogue *srrm3*, which is responsible for microexon inclusion in photoreceptor transcripts in zebrafish (Ciampi et al., 2022 [↗](#)). A vertebrate model is necessary for these studies; although there is a similar splicing mechanism in *Drosophila melanogaster*, there is no overlap of the individual microexons (Torres-Mendez et al., 2021 [↗](#)). Thus, we used larval zebrafish, a vertebrate that supports more high-throughput studies than mammalian models (Thyme et al., 2019 [↗](#)), to assess the brain activity, brain structure, and behavior of zebrafish larvae mutant for 45 microexons (**Figure 1A** [↗](#)).

Results

A recently developed computational tool identified microexons that are differentially spliced during mouse brain development and conserved in zebrafish (Parada et al., 2021 [↗](#)). Based on the intersection of these two sets, 95 microexons fulfilled both parameters (Table S1). Of these 95 microexons, 42 exist in a canonical layout in the zebrafish genome, with both a UGC and UC repeat – or similar polypyrimidine tract – directly upstream of the alternatively spliced exon (Gonatosopoulos-Pournatzis et al., 2018 [↗](#)) (Table S1, Figure S1), indicating that *Srrm4* likely controls their inclusion. Of the remaining microexons, 44 are organized similarly to the canonical layout, typically with either a UGC or UC repeat. Thus, they may also be regulated by *Srrm4*. The protein sequences of the 95 microexons are highly conserved between mouse and zebrafish (**Figure 1B** [↗](#)), more so than the full-length protein sequences (**Figure 1C** [↗](#)). The genes containing microexons are enriched for those involved in neuronal development as well as the cytoskeleton (**Figure 1D** [↗](#)). The majority of these microexons are included in transcripts as the brain develops, rather than lost, in both mouse and zebrafish (**Figure 1E** [↗](#)), making CRISPR/Cas9 removal of the microexon an ideal approach for studying the function of the added amino acids.

Using CRISPR/Cas9, we generated lines that removed 45 conserved microexons (Table S2) and assayed larval brain activity, brain structure, and behavior (**Figure 1A** [↗](#)). Four guide RNAs were used, two on each side of the microexon (Table S2, Figure S1). For microexons with upstream regulatory elements that are likely important for splicing, these elements were also removed (Figure S1). While we were most interested in those microexons only present after 24 hours post-fertilization (**Figure 1E** [↗](#)), as this is when the brain is forming, we also removed microexons from several genes with an earlier expression of the isoform. For eight mutant lines, we confirmed that the microexon was eliminated from the transcripts as expected (Figure S2). Although our genomic deletion did not yield unexpected isoforms, qRT-PCR on these eight lines revealed significant downregulation for the homozygous *vav2* mutant (Figure S2), indicating possibly complex genetic regulation. Protein-truncating mutations in eleven additional genes that contain microexons revealed developmental and neural phenotypes in zebrafish (Figure S3, Figure S4), indicating that the genes themselves are involved in biologically relevant pathways. Three of these genes – *tenm4*, *sptan1*, and *ppp6r3* – are also in our microexon line collection.

To uncover neural phenotypes, we assessed baseline and stimulus-driven movement. Homozygous mutant and control sibling larvae were subjected to a behavioral pipeline in 96-well plates from 4–7 days post-fertilization (dpf) (**Figure 2A** [↗](#)). Only a small number of strong, repeatable phenotypes were observed either at baseline or in response to acoustic or visual stimuli (**Figure 2B** [↗](#), **Figure 2C** [↗](#), Figure S5, Figure S6, Figure S7). Baseline behavioral differences were observed in multiple movement categories. For example, *rapgef2* mutants showed an increased movement frequency (**Figure 2B** [↗](#), **Figure 2D** [↗](#)) but no changes to the characteristics of these movements (magnitude) or location in the well. In contrast, *dop1a* mutants had reduced daytime movement and an increased well-edge preference specifically at night (Figure S6, Figure S8, **Figure 2B** [↗](#),

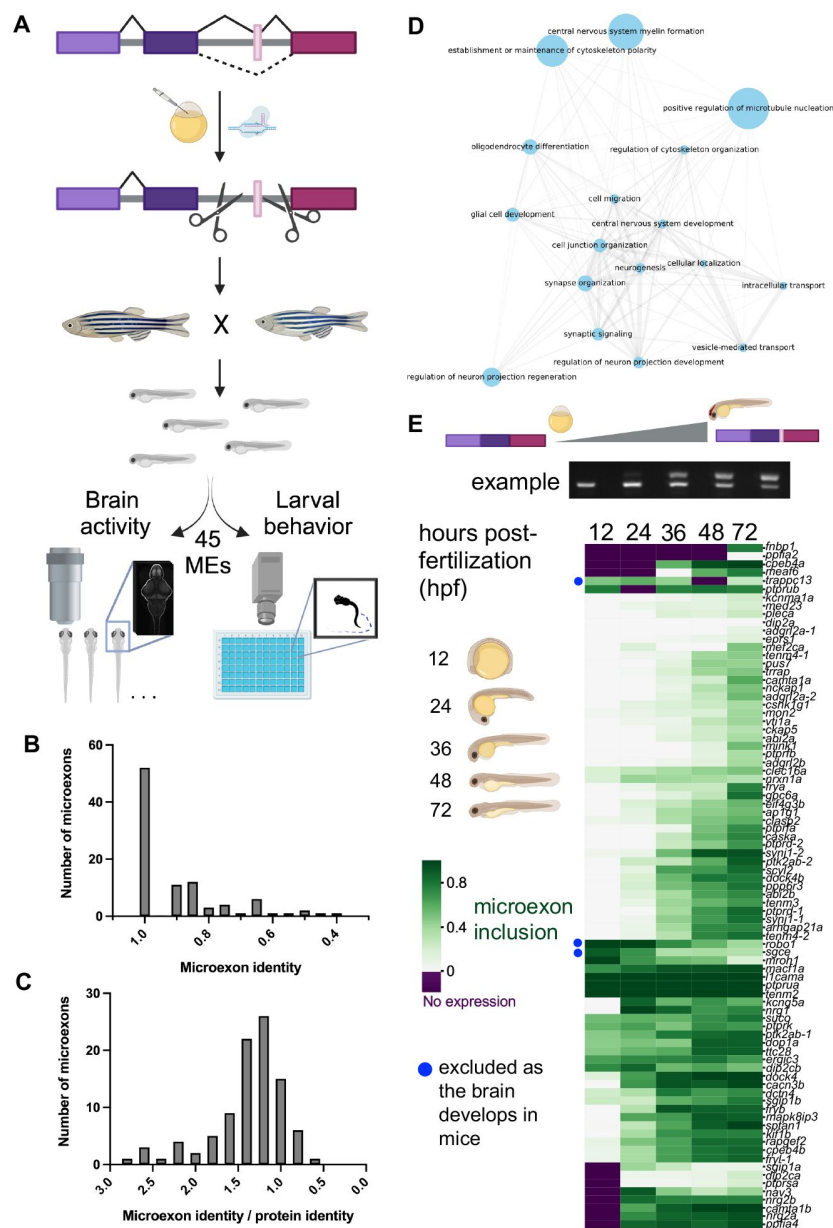


Figure 1.

Generation and analysis of zebrafish with mutations that remove conserved, developmentally regulated microexons.

(A) Pipeline of the screen. Mutant lines with alternatively spliced microexons removed were generated with CRISPR/Cas9, crossed together, and sibling larvae were assessed for changes to brain morphology, brain activity, and behavioral profiling. Created with BioRender.com. (B) The amino acid sequence identity of 95 zebrafish microexons compared to mouse. (C) The amino acid sequence identity of 95 zebrafish microexons compared to mouse and divided by the sequence identity of the entire protein. (D) Gene Ontology analysis of biological processes associated with the 95 mouse microexons that are conserved in zebrafish. The analysis was completed using the PANTHER classification system. (E) Quantification of reverse transcription PCR (RT-PCR) for microexon-containing regions over zebrafish development. These data were clustered using the default seaborn clustermap settings (method = 'average'). Panels A and E were created using BioRender.com.

Figure 2D [↗](#)). Other baseline movement phenotypes included a well-center preference for *vav2* mutants, a decreased movement magnitude for *ptprd-2* (also referred to in the literature as meA) mutants (Figure S8), and an increased movement magnitude for *ptprd-1* (also referred to in the literature as meB) mutants (**Figure 2D** [↗](#)). Dark-flash response phenotypes included a reduced latency for *elif4g3b* mutants and a reduced response frequency for *ptprd-1* (meB) and *dop1a* mutants (**Figure 2C** [↗](#), **Figure 2D** [↗](#), Figure S8). Acoustic response phenotypes included an increased frequency of response to strong stimuli for *ppp6r3* mutants and a reduced frequency of response to strong acoustic stimuli for *ptprd-2*.

To detect changes to brain development and function, we collected phospho-Erk (pErk) brain activity maps under unstimulated conditions at 6 dpf. Large morphological differences were identified using the Jacobian matrix derived from the image registration process (Jefferis et al., 2007 [↗](#); Rohlfing and Maurer, 2003 [↗](#)). Similar to the behavioral findings, few phenotypes were observed in brain activity (**Figure 3A** [↗](#), Figure S9, Figure S10) or brain structure (**Figure 3B** [↗](#), Figure S11, Figure S12) in homozygous mutants compared to wild-type control siblings when quantified by region (**Figure 3C** [↗](#)). Repeatable phenotypes were mild (**Figure 3D** [↗](#)) compared to a mutant with a moderate phenotype and both increased and decreased activity and structure (see control-*tcf7l2*) (Capps et al., 2024 [↗](#)). Only one mutant, *ppp6r3*, displayed substantial and repeatable differences in brain morphology (**Figure 3B** [↗](#), **Figure 3E** [↗](#)), with reduced size observed mainly in the thalamus and neighboring pretectum. While the loss of both copies of microexons was required for most phenotypes, brain activity differences were observed in *vti1a* and *sptan1* heterozygotes (**Figure 3F** [↗](#)) as well as *vav2* and *mapk8ip3* (Figure S9, Figure S10). While repeatable phenotypes in brain activity and structure were observed in a small subset of mutants, other zebrafish studies using identical methods have typically uncovered brain activity phenotypes for approximately half of the lines that are stronger than any we observed (**Figure 3G** [↗](#)) (Capps et al., 2024 [↗](#); Thyme et al., 2019 [↗](#); Weinschutz Mendes et al., 2023 [↗](#)).

Discussion

Our study provides a broad overview of the larval zebrafish neural phenotypes resulting from removing developmentally regulated microexons. Here, we leveraged our previously successful brain activity mapping and behavioral profiling strategy, which has high sensitivity and has uncovered phenotypes for dozens of mutants (Thyme et al., 2019 [↗](#)). While only a few zebrafish lines had neural phenotypes, most of the microexons with behavioral or brain activity differences were previously unstudied and found in diverse protein classes.

The small number of observed phenotypes was unexpected based on conservation of the microexons (**Figure 1B** [↗](#), **Figure 1C** [↗](#)) and the discovery rates of previous screens using similar methods (Capps et al., 2024 [↗](#); Thyme et al., 2019 [↗](#); Weinschutz Mendes et al., 2023 [↗](#)). Although most of these studies focused on protein-truncating alleles, we have found phenotypes in lines with missense mutations (Capps et al., 2024 [↗](#)). However, this outcome is consistent with a recent, similar study of microexons in zebrafish (Lopez-Blanch et al., 2024 [↗](#)). Although only three of the 18 microexons this group studied are in the same genes as our set (*ap1g1*, *vav2*, and *vti1a*), the phenotypes for individual mutants are similar and this work shares the same overall finding of minimal phenotypes in zebrafish larvae. Importantly, the pErk brain activity mapping method we used is highly sensitive, significantly minimizing the likelihood that substantial zebrafish larval phenotypes from individual microexon removal are being missed. In our published work (Capps et al., 2024 [↗](#); Thyme et al., 2019 [↗](#)), we showed that brain activity can be drastically impacted without manifesting in differences in those behaviors assessed in a typical larval screen such as was completed by Lopez-Blanch et al. Furthermore, the genes that contain microexons are developmentally important (Figure S3, Figure S4). For instance, the *caska* loss-of-function mutant

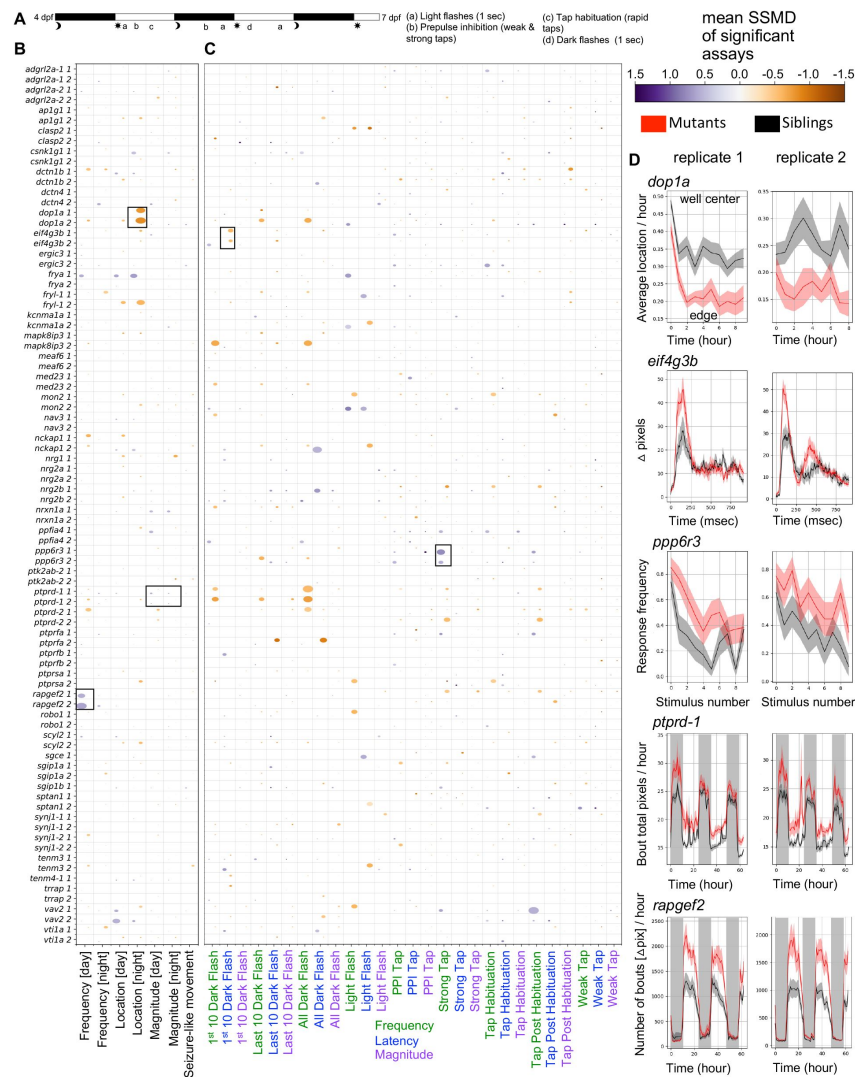


Figure 2.

Larval behavioral phenotypes of zebrafish with microexons removed.

(A) Summary of behavioral pipeline. (B) Baseline behavioral phenotypes for microexon mutants. The labels “1” and “2” indicate biological replicates. The data shown is for homozygous mutant larvae compared to wild-type siblings. Comparisons to the heterozygous larvae are removed for clarity and available in the Supplementary Materials, as they often have even milder phenotypes than homozygous. The size of the bubble represents the percent of significant measurements in the summarized category, and the color represents the mean of the strictly standardized mean difference (SSMD) of the significant assays in that category. (C) Stimulus-driven behavioral phenotypes for microexon mutants. The labels “1” and “2” indicate biological replicates. The bubble size and color are calculated the same as in panel B. (D) Examples of behavioral phenotypes. The black boxes in panels B and C correspond to the selected plots. Wild-type siblings (black) are compared to the homozygous (red), and the plots show mean $\pm$ s.e.m.. All N are in Table S2. Kruskal-Wallis ANOVA p-values for the selected plots are as follows: *dop1a* center preference during the first night (day0night_boutcenterfraction_3600) = 0.00006/0.0008, N +/+ = 15 (run 1) and 14 (run 2), N -/- = 17 (run 1) and 15 (run 2); *eif4g3b* p-values are not calculated for response traces (shown is all dark flashes in block 1), p-values for the latency for the first 10 dark flashes in block 1 (day6dpdf1a_responselatency) = 0.026/0.0008, N +/+ = 16 (run 1) and 15 (run 2), N -/- = 15 (run 1) and 21 (run 2); *ppp6r3* frequency of response to strong acoustic stimuli with a sound frequency of 1000 hertz that precede the habituation block (day5dpfhab1pre_responsefrequency_1a1f1000d5p) = 0.002/0.016, N +/+ = 19 (run 1) and 20 (run 2), N -/- = 21 (run 1) and 29 (run 2); *ptprd-1* pixels moved in each bout for the duration of the experiment (combo_boutcumulativemovement_3600) = 0.001/0.00002, N +/+ = 22 (run 1) and 21 (run 2), N -/- = 26 (run 1) and 18 (run 2); *rapgef2* number of bouts for the duration of the experiment calculated using the delta pixel data in each frame (dpix_numberofbouts_3600) = 0.002/0.001, N +/+ = 21 (run 1) and 23 (run 2), N -/- = 21 (run 1) and 20 (run 2).

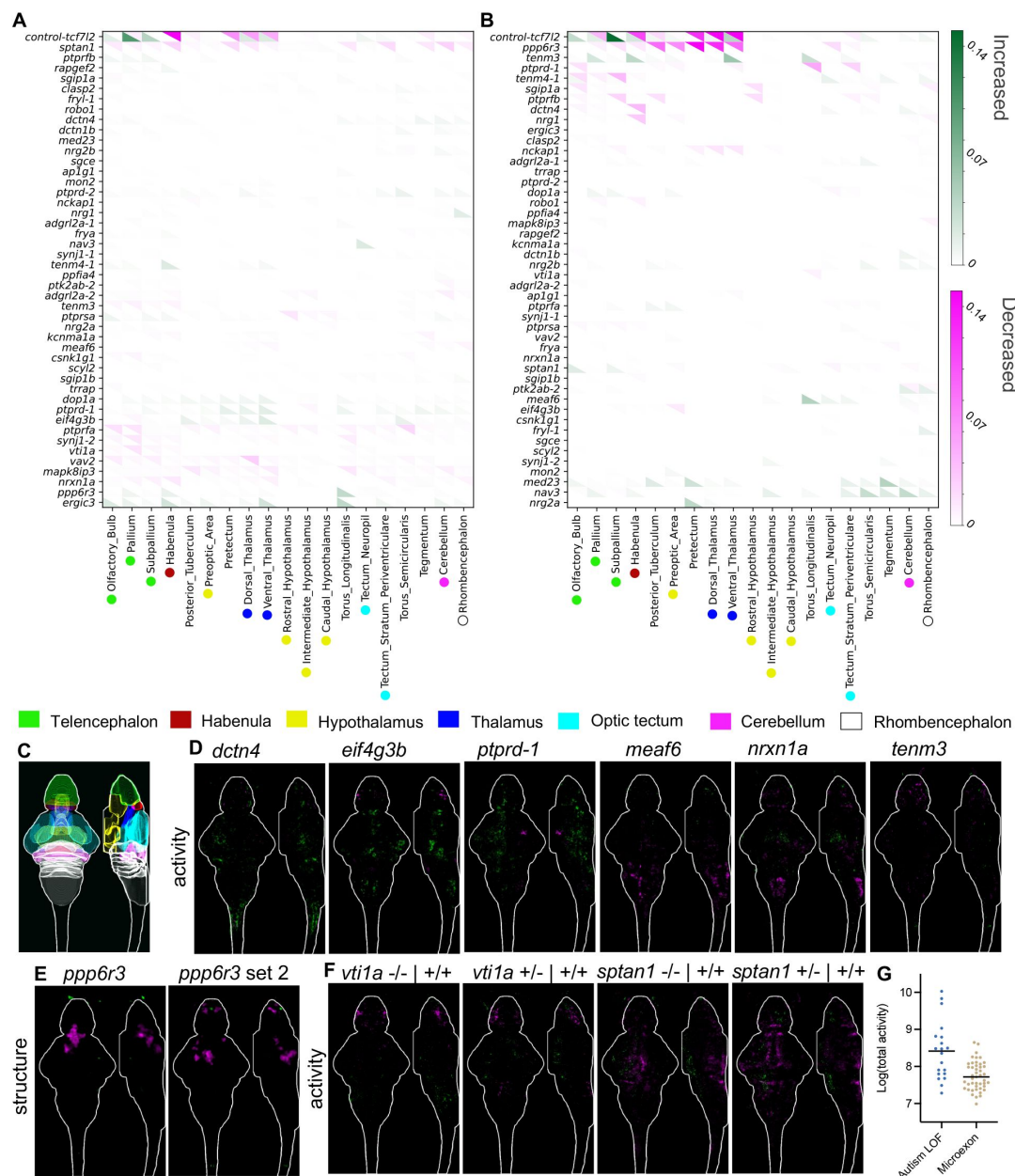


Figure 3.

Whole-brain activity and morphology phenotypes of zebrafish with microexons removed.

(A) Clustered summary of pErk comparisons between homozygous mutants and wild-type control siblings, where magenta represents decreased activity and green represents increased. The signal in each region was summed and divided by the region size. The N for all experiments is available in Table S2. (B) Clustered summary of structure comparisons between homozygous mutants and wild-type control siblings, where magenta represents decreased size and green represents increased. (C) Location of major regions in the zebrafish brain based on masks from the Z-Brain atlas (Randlett et al., 2015). (D) Brain imaging for microexon mutants with repeatable brain activity phenotypes. The brain images represent the significant signal difference between homozygous and wild-type control siblings. They are shown as sum-of-slices projections (Z- and X-axes) with the white outline representing the zebrafish brain. (E) Structural phenotype of the *ppp6r3* mutant, with replicates shown side-by-side. The magenta indicates decreased size. (F) Brain imaging for two microexon mutants with brain activity phenotypes that are similar for both the heterozygous and homozygous mutants. (G) Comparison of the total brain activity signal between homozygous microexon mutants from this work and mutants for autism risk genes from (Capps et al., 2024). Both increased and decreased activity are considered and a single comparison rather than the average of the repeats.

displayed substantial changes to brain activity, a smaller brain size, and multiple behavioral phenotypes. Loss-of-function mutants for *sptan1* and *ppp6r3* are stronger than lines that only remove the microexon (Figure S3, Figure S4).

The limited effects of individual microexon deletions suggest that stronger phenotypes may require disrupting multiple microexons or examining later stages of development. It is possible that perturbation of multiple microexons simultaneously, as was observed in the brains of autistic individuals (Irimia et al., 2014), is necessary to produce pronounced effects on early neurodevelopment. Zebrafish homozygous for *srrm4* deletion did not display any overt developmental phenotypes (Ciampi et al., 2022), in contrast to a previous morpholino study (Calarco et al., 2009). More recently, it was revealed that loss of *srrm4* has minimal impacts on behavior (Lopez-Blanch et al., 2024) and brain structure (Gupta et al., 2025), while *srrm3* mutants have significant neural phenotypes and early mortality (Ciampi et al., 2022; Lopez-Blanch et al., 2024). This finding indicates that *Srrm3* loss is necessary for complete elimination of these microexons, but studying microexons in the context of this line is also complicated by possible downstream consequences of severe vision loss. Alternatively, although the microexons are present at larval stages, they may only become important during later stages of neuronal maturation that support more complex behaviors. Mouse models were studied as adults and for disruptions to social interaction, learning, and memory (Gonatopoulos-Pournatzis et al., 2020; Han et al., 2024), which larvae are not yet capable of (Dreosti et al., 2015; Valente et al., 2012).

Our study extends on findings for previously studied genes containing microexons. We discovered increased brain activity (Figure 3D) and an altered dark flash response in mutants for the *elf4g3b* (Figure 2D), whereas a mouse model with the microexon removed was described as having no behavior phenotypes (Gonatopoulos-Pournatzis et al., 2020). That study, however, focused on social behavior, learning, and memory and may not have included stimuli analogous to dark flashes. Alternatively, microexon removal in zebrafish could have differential behavioral impacts than in rodents. Mice mutant for the *ptprd-2* (meA) microexon have increased movement and interrupted sleep (Park et al., 2020). Both zebrafish mutants in *ptprd* demonstrate the opposite, with reduced daytime movement and increased sleep during the day for *ptprd-2* (Figure S8). While the underlying disrupted circuitry could differ between the species, the daytime sleep increase raises the possibility of a rebound response to undetected nighttime sleep disruptions. In other cases, the gene that contains the microexon has been extensively studied, but research on the microexon itself is absent. For example, the function of SNARE protein *Vti1a* in neural development and dense core vesicle biogenesis has been studied in great depth (Bollmann et al., 2022; Emperador-Melero et al., 2018; Kunwar et al., 2011; Sokpor et al., 2021; Walter et al., 2014). Its brain-specific microexon, however, was recognized over twenty years ago (Antonin et al., 2000), and its specific importance has been largely ignored. We discovered reduced brain activity mainly in the telencephalon in both homozygous and heterozygous lines compared to wild-type siblings (Figure 3F), highlighting the relevance of considering multiple conserved isoforms when characterizing protein function in the brain.

Mutants for unstudied microexons with the strongest phenotypes come from diverse protein classes (Gonatopoulos-Pournatzis and Blencowe, 2020), indicating this splicing program impacts neurodevelopmental processes beyond *trans*-synaptic partner selection. The only mutant with a substantial brain size difference was *ppp6r3*, which encodes a regulator of protein phosphatase 6 (PP6), and this reduced size was present and more severe in the predicted protein-truncating *ppp6r3^{sa16892}* line (Figure S3). Although little is known about this protein, it has been linked to cell cycle (Heo et al., 2020) and cell death (Wu et al., 2022), nominating hypotheses for determining future research on the source of the reduced size. *Dctn4* is part of the dynactin complex, *Mapk8ip3* is involved in Kinesin-1-Dependent axonal trafficking (Platzer et al., 2019; Watt et al., 2015), and *Sptan1* is a cytoskeleton scaffold protein (Huang et al., 2017); the number microexon-containing genes related to the cytoskeleton (Figure 1D) and also trafficking

(e.g., *vti1a*), suggests that microexons may be important to the specialized cytoskeletal needs of developing neurons. The functions of microexons in cytoskeletal proteins are beginning to be discovered (Poliński et al., 2023 [DOI](#)). Transcriptional regulation is represented by the *meaf6* mutant, while *Rapgef2*, which has the strongest behavioral phenotypes upon microexon removal (Figure 3D [DOI](#)), is a signaling protein.

Although our list is not comprehensive, as many microexons remain unstudied, we have prioritized several microexons for future study from the 45 tested. For seven mutants, we confirmed that the microexon was cleanly removed without unanticipated effects on isoforms or transcript levels (Figure S2), including for some of the most interesting that are entirely unstudied (*ppp6r3*, *sptan1*, *meaf6*). For those with phenotypes that are not yet confirmed by qRT-PCR (*rapgef2*, *dctn4*, *dop1a*, *mapk8ip3*), studies of gene loss-of-function in zebrafish and mice reveal far stronger phenotypes and lethality, suggesting that our lines impact only the microexon (Abeler-Dörner et al., 2020 [DOI](#); Drerup and Nechiporuk, 2013 [DOI](#); Satyanarayana et al., 2010 [DOI](#); Tuttle et al., 2019 [DOI](#)). In future work, proximity-dependent ligation in zebrafish lines with and without the microexon could reveal differential binding partners *in vivo* (Rosenthal et al., 2021 [DOI](#)), as microexons are often found in surface-accessible protein domains and can regulate protein-protein interactions (Dergai et al., 2010 [DOI](#); Irimia et al., 2014 [DOI](#)). The discovery of microexon mutants with neural phenotypes lays the groundwork for future investigation of impacted neurodevelopmental pathways and how interactomes differ between isoforms.

Materials and Methods

Zebrafish husbandry

Zebrafish experiments were approved by the UAB Institutional Animal Care and Use Committee (IACUC protocols 22155 and 21744) and UMass Chan Institutional Animal Care and Use Committee (IACUC protocol 202300000053). Mutants were generated in an Ekkwill-based strain using CRISPR/Cas9 as previously described (Capps et al., 2024 [DOI](#); Thyme et al., 2019 [DOI](#)) (Table S2). Both adults and larvae were maintained on a 14 h/10 h light/dark cycle at 28°C. Experimental larvae were maintained at a density of less than 160 per dish in 150 mm petri dishes in fish water with methylene blue, and debris was removed at least twice prior to experimentation. Visibly unhealthy larvae and those without inflated swim bladders were excluded from experiments. Control animals were always siblings from the same clutch and derived from a single parental pair, and all larvae were genotyped after experimentation. Even with using sibling controls and collecting multiple biological replicates from individual parents, the possibility remains that linked genetic variation may have contributed to the mild phenotypes we observed, as only a single line was generated.

Brain activity and morphology

Phosphorylated-ERK (pErk) antibody staining was conducted as previously described (Capps et al., 2024 [DOI](#); Thyme et al., 2019 [DOI](#)). The total ERK antibody (Cell Signaling, #4696) was used at 3:1000 and pErk antibody was used at 1:500 (Cell Signaling, #4370). Typically, the primary antibody exposure was for 2-3 days and secondary for one day. Image stacks were collected with a Zeiss LSM 900 upright confocal microscope using a 20x/1.0 NA water-dipping objective. These stacks were registered to the standard zebrafish Z-Brain reference using Computational Morphometry Toolkit (CMTK) (Jefferis et al., 2007 [DOI](#); Randlett et al., 2015 [DOI](#); Rohlfing and Maurer, 2003 [DOI](#); Thyme et al., 2019 [DOI](#)). The significance threshold for MapMAPPING (Randlett et al., 2015 [DOI](#); Thyme et al., 2019 [DOI](#)) was set based on a false discovery rate (FDR) where 0.05% of control pixels would be called as significant for both the brain activity and structural measurements.

Larval behavior

Larval behavior assays were conducted and analyzed as previously described (Capps et al., 2024 [↗](#); Joo et al., 2020 [↗](#); Thyme et al., 2019 [↗](#)). The pipeline begins on the evening of larval stage 4 dpf and includes the following stimuli: 5 dpf light flashes (9:11-9:25), mixed acoustic stimuli (prepulse, strong, and weak, from 9:38-2:59), and three blocks of acoustic habituation (3:35-6:35); the night between 5 and 6 dpf mixed acoustic stimuli (1:02-5:00) and light flashes (6:01-6:20); 6 dpf three blocks of dark flashes with an hour between them (10:00-3:00) and mixed acoustic stimuli and light flashes (4:02-6:00). Stimulus responses were quantified from high-speed (285 frames-per-second) 1-second-long movies. All code for behavior analysis is available at <https://github.com/thymelab/ZebrafishBehavior> [↗](#). Frequency of movement, magnitude of movement, and location preferences were calculated for the baseline data. An example of a “frequency” measure would be the active seconds / hour. An example of the “magnitude” measure would be the movement velocity of a bout. An example of a “location” measure would be the fraction of the bout time spent in the center of the well. Frequency, response magnitude parameters, and latency to response are also calculated for each stimulus response from the high-speed movies. The stimulus responses are also broken up into subsets, such as the dark flashes that occur at the beginning of the dark flash block and those that occur at the end. Multiple measures are calculated for the experiment, and both summarized and raw measures are shown in **Figure 2** [↗](#) and the Supplementary Material. The summarized data is generated by merging related terms (e.g., all the “frequency” measures) together. The size of the bubble represents the percent of significant measurements in the summarized category, and the color represents the mean of the strictly standardized mean difference (SSMD) of the significant assays in that category. It is possible for some measures in the merged group to be increased (purple) and some decreased (orange), such as if a behavioral change occurs between 4 and 6 dpf, which is why it is possible to have two offset bubbles in each square. The scripts for merging these measurements and generating the summarized bubble plot graphs are available at <https://github.com/thymelab/DownstreamAnalysis> [↗](#).

Reverse Transcription PCR (RT-PCR and qRT-PCR)

RNA was extracted from zebrafish embryos using the E.Z.N.A. MicroElute Total RNA Kit (Omega Bio-Tek R6834-02), and cDNA synthesis was performed with iScript Reverse Transcription Supermix (Bio-Rad #1708840). Following PCR with GoTaq polymerase (Promega M7123) and primers (Table S1), samples were run on a 4% agarose gel and the band intensities were quantified using Fiji (Schindelin et al., 2012 [↗](#)). For qRT-PCR, we combined 3-4 zebrafish heads at 6 dpf for each biological replicate. For RT-PCR, these separate biological samples were combined into one per genotype. We homogenized the heads and performed RNA extraction and cDNA synthesis as described above. We then performed qRT-PCR with SsoAdvanced Universal SYBR Green Supermix (Bio-Rad #1725270EDU) according to the manufacturer’s protocol and using primers in Table S1.

Data and materials availability

The mutants described in the paper are available from ZIRC. Code is available from <https://github.com/thymelab> [↗](#). Processed behavioral and imaging data is available from Zenodo under the following DOIs: 10.5281/zenodo.13137486 (behavior output files), 10.5281/zenodo.13138646 (behavior input files), and 10.5281/zenodo.13138766 (brain activity mapping stacks).

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Additional information

Author contributions

SBT conceived of the study, analyzed imaging and behavioral data, and wrote the manuscript. VM, JW, and CLC generated the mutant lines. CCSC and KM initiated the project and CCSC collected most of the experimental data. MESC, WCG, MCK, CLC, and EGT contributed to the experimental data in this work by genotyping and propagating zebrafish lines, running larval behavior experiments, and collecting brain imaging data.

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Additional files

Table S1 [↗](#)

Table S2 [↗](#)

Supplementary Figures S1-S12 [↗](#)

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Reviewer #1 (Public review):

Summary:

The authors use high-throughput gene editing technology in larval zebrafish to address whether microexons play important roles in the development and functional output of larval circuits. They find that individual microexon deletions rarely impact behavior, brain morphology, or activity, and raise the possibility that behavioral dysregulation occurs only with more global loss of microexon splicing regulation. Other possibilities exist: perhaps microexon splicing is more critical for later stages of brain development, perhaps microexon splicing is more critical in mammals, or perhaps the behavioral phenotypes observed when microexon splicing is lost are associated with loss of splicing in only a few genes.

Strengths:

- The authors provide a qualitative analysis of microexon inclusion during early zebrafish development
- The authors provide comprehensive phenotyping of microexon mutants, addressing the role of individual microexons in the regulation of brain morphology, activity, and behavior.

<https://doi.org/10.7554/eLife.101790.2.sa2>

Reviewer #3 (Public review):

Summary:

This paper sought to understand how microexons influence early brain function. By selectively deleting a large number of conserved microexons and then phenotyping the mutants with a behavior and brain activity assays, the authors find that most microexons have minimal effects on the global brain activity and broad behaviors of the larval fish--although a few do have phenotypes.

Strengths:

The work takes full advantage of the scale that is afforded in zebrafish, generating a large mutant collection that is missing microexons and systematically phenotyping them with high throughput behaviour and brain activity assays. The work lays an important foundation for future studies that seek to uncover the likely subtle roles that single microexons will play in shaping development and behavior.

Weaknesses:

Although the manuscript includes evidence for many mutants that microexon deletion has minimal effect on full length transcript levels, some of the microexon loss does alter transcript levels. Since the mutations usually yielded no phenotype, these effects on full-length transcripts are unlikely to be a major confound. For microexon mutants displaying phenotypes, future work will have to tease apart whether secondary effects on the transcripts are contributing to the phenotype.

<https://doi.org/10.7554/eLife.101790.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The authors use high-throughput gene editing technology in larval zebrafish to address whether microexons play important roles in the development and functional output of larval circuits. They find that individual microexon deletions rarely impact behavior, brain morphology, or activity, and raise the possibility that behavioral dysregulation occurs only with more global loss of microexon splicing regulation. Other possibilities exist: perhaps microexon splicing is more critical for later stages of brain development, perhaps microexon splicing is more critical in mammals, or perhaps the behavioral phenotypes observed when microexon splicing is lost are associated with loss of splicing in only a few genes.

A few questions remain:

(1) What is the behavioral consequence for loss of srrm4 and/or loss-of-function mutations in other genes encoding microexon splicing machinery in zebrafish?

It has been established that srrm4 mutants exhibit no overt morphological phenotypes and are not visually impaired (Ciampi et al., 2022). We are coordinating our publication with Lopez-Blanch et al. (<https://doi.org/10.1101/2024.10.23.619860>), which shows that srrm4 mutants also have minimal behavioral phenotypes. In contrast, srrm3 mutants have severe vision loss, early mortality, and numerous neural and behavioral phenotypes (Ciampi et al., 2022; Lopez-Blanch et al., 2024). We now point out the phenotypes of srrm3/srrm4 mutants in the manuscript.

We chose not to generate and characterize the behavior and brain activity of srrm3/srrm4 mutants for two reasons: 1) we were aware of two other labs in the zebrafish community that had generated srrm3 and/or srrm4 mutants (Ciampi et al., 2022 and Gupta et al., 2024, <https://doi.org/10.1101/2024.11.29.626094>; Lopez-Blanch et al., 2024, <https://doi.org/10.1101/2024.10.23.619860>), and 2) we were far more interested in determining the importance of individual microexons to protein function, rather than loss of the entire splicing program. Microexon inclusion can be controlled by different splicing regulators, such as srrm3 (Ciampi et al., 2022) and possibly other unknown factors. Genetic compensation in srrm4 mutants could also result in microexons still being included through actions of other splicing regulators, complicating the analysis of these regulators. We mention srrm4 in the manuscript to point out that some selected microexons are adjacent to regulatory elements expected of this pathway. We did not, however, choose microexons to mutate based on whether they were

regulated by Srrm4, making the characterization of srrm3/srrm4 mutants disconnected from our overarching project goal.

We have edited the Introduction as follows to clarify our goal: “Studies of splicing regulators such as srrm4 impact the entire splicing program, making it impossible to determine the importance of individual microexons to protein function. Further, microexons could still be differentially included in a regulatory mutant via compensation by other splicing factors ...”

(2) What is the consequence of loss-of-function in microexon splicing genes on splicing of the genes studied (especially those for which phenotypes were observed).

We are unclear whether “microexon splicing genes” refers to the splicing regulators srrm3/srrm4, which we choose not to study in this work (see response to point #1 above), or the genes that contain microexons. The severe visual phenotypes of srrm3 mutants confounds the study of microexon splicing in this line because altered splicing levels could be due to downstream changes in this significantly different developmental context. A detailed discussion of splicing consequences on removal of microexons from microexoncontaining genes is in the response to point #4 below.

(3) For the microexons whose loss is associated with substantial behavioral, morphological, or activity changes, are the same changes observed in loss-of-function mutants for these genes?

In the first version of the manuscript, we had included two explicit comparisons of microexon loss with a standard loss-of-function allele, one with a phenotype and one without, in Figure S1 (now Figures S3 and S4) of this manuscript. Beyond the two pairs we had included, Lopez-Blanch et al. (<https://doi.org/10.1101/2024.10.23.619860>) described mild behavioral phenotypes for a microexon removal for kif1b, and we showed developmental abnormalities for the kif1b loss-of-function allele (now Figure S3). We have now added a predicted protein-truncating allele for ppp6r3. This new line has phenotypes that are similar but slightly stronger in brain activity and structure than the mutant that lacks only the microexon. The prior Figure S1 (now Figures S3 and S4) was only briefly mentioned in the first version of the manuscript, and we now clarify this point in the Results: “Protein-truncating mutations in eleven additional genes that contain microexons revealed developmental and neural phenotypes in zebrafish (Figure S3, Figure S4), indicating that the genes themselves are involved in biologically relevant pathways. Three of these genes—tenm4, sptan1, and ppp6r3 – are also in our microexon line collection.”

Additionally, we can draw expected conclusions from the literature, as some genes with our microexon mutations have been studied as typical mutants in zebrafish or mice. We have modified our manuscript to include a discussion of both loss-of-function zebrafish and mouse mutants. See the response to below point #4.

(4) Do “microexon mutations” presented here result in the precise loss of those microexons from the mRNA sequence? E.g. are there other impacts on mRNA sequence or abundance?

We acknowledge that unexpected changes to the mRNA of the tested mutants could occur following microexon removal. In particular, all regulatory elements should be removed from the region surrounding the microexon, as any remaining elements could drive the inclusion of unexpected exons that result in premature stop codons.

First, we have clarified our generated mutant alleles by adding a figure (Figure S1) that details the location of the gRNA cut sites in relation to the microexon, its predicted regulatory elements, and its neighboring exons.

Second, we have experimentally determined whether the mRNA was modified as expected for a subset of mutants with phenotypes. In all eight tested lines (Figure S2), the microexon was precisely eliminated without causing any other effects on the sequence of the transcript in the neighboring region. We did, however, observe an effect on transcript abundance for one homozygous mutant (*vav2*). It is possible that complex forms of genetic regulation are occurring that are not induced by unexpected isoforms or premature stop codons. Interestingly, Lopez-Blanch et al. (<https://doi.org/10.1101/2024.10.23.619860>) eliminated a different microexon in *vav2* and also observed a subtle well center preference. If their allele from an entirely different intronic region also results in transcript downregulation, it would support the hypothesis of genetic compensation through atypical pathways. If not, it is likely this phenotype is due specifically to removal of the microexon protein sequence. Not all mutants with phenotypes could be assessed with qRT-PCR because some were no longer present in the lab. All lines were generated in a similar way, however, removing both the microexon and neighboring regulatory elements while avoiding the neighboring exons. Accordingly, we now also explicitly point out those where the clean loss of the microexon was confirmed (*EIF4G3B*, *PPP6R3*, *SPTAN1*, *VTI1A*, *MEAF6*, *NRXN1A*, *TENM3*) and those with possibly interesting phenotypes that were not confirmed (*PTPRD-1*, *PTPRD-2*, *RAPGEF2*, *DCTN4*, *DOP1A*, *MAPK8IP3*).

Third, we have further emphasized in the manuscript that these observed phenotypes are extremely mild compared to those observed in over one hundred protein-truncating mutations we have assessed in previous (Thyme et al., 2019; Capps et al., 2024) and unpublished ongoing work. We showed data for one mutant, *tcf7l2*, which we consider to have moderately strong neural phenotypes, and we have extended this comparison in the revision (new Figure 3G). Additionally, loss-of-function alleles for some microexon-containing genes have strong developmental phenotypes, as we showed in Figure S1 (now Figures S3 and S4) of this manuscript in addition to our published work (Thyme et al., 2019; Capps et al., 2024). It is known from the literature that the loss-of-function mutants for *MAPK8IP3* are stronger than we observed here (Tuttle, et al., 2019), suggesting that only the microexon is removed in our line. The microexons in *PTPRD* are also well-studied in mice, and we expect that only the microexon was removed in our lines. Both *DCTN4* and *RAPGEF2* are completely lethal prior to weaning in mice (the International Mouse Phenotyping Consortium).

(5) Microexons with a "canonical layout" (containing TGC / UC repeats) were selected based on the likelihood that they are regulated by SRRM4. Are there other parallel pathways important for regulating the inclusion of microexons? Is it possible to speculate on whether they might be more important in zebrafish or in the case of early brain development?

The microexons were not selected based on the likelihood that they were regulated by *SRRM4*. We have clarified the manuscript regarding this point. There are parallel pathways that can control the inclusion of microexons, such as *SRRM3* (Ciampi et al., 2022). It is wellknown that loss of *srrm3* has a stronger impact on zebrafish development than *srrm4* (Ciampi et al., 2022). The goal of our work was not to investigate these splicing regulators but instead to determine the individual importance of these highly conserved protein changes.

Strengths:

(1) The authors provide a qualitative analysis of splicing plasticity for microexons during early zebrafish development.

(2) The authors provide comprehensive phenotyping of microexon mutants, addressing the role of individual microexons in the regulation of brain morphology, activity, and behavior.

We thank the reviewer for their support. The pErk brain activity mapping method is highly sensitive, significantly minimizing the likelihood that the field has simply not looked hard enough for a neural phenotype in these microexon mutants. In our published work (Thyme et al., 2019), we show that brain activity can be drastically impacted without manifesting in differences in those behaviors assessed in a typical larval screen (e.g., *tcf4*, *cnnm2*, and more).

Weaknesses:

*(1) It is difficult to interpret the largely negative findings reported in this paper without knowing how the loss of *srrm4* affects brain activity, morphology, and behavior in zebrafish.*

See response to point 1.

(2) The authors do not present experiments directly testing the effects of their mutations on RNA splicing/abundance.

See response to point 4.

(3) A comparison between loss-of-function phenotypes and loss-of-microexon splicing phenotypes could help interpret the findings from positive hits.

See response to points 3 and 4.

Reviewer #2 (Public review):

Summary:

The manuscript from Calhoun et al. uses a well-established screening protocol to investigate the functions of microexons in zebrafish neurodevelopment. Microexons have gained prominence recently due to their enriched expression in neural tissues and misregulation in autism spectrum disease. However, screening of microexon functionality has thus far been limited in scope. The authors address this lack of knowledge by establishing zebrafish microexon CRISPR deletion lines for 45 microexons chosen in genes likely to play a role in CNS development. Using their high throughput protocol to test larval behaviour, brain activity, and brain structure, a modest group of 9 deletion lines was revealed to have neurodevelopmental functions, including 2 previously known to be functionally important.

Strengths:

(1) This work advances the state of knowledge in the microexon field and represents a starting point for future detailed investigations of the function of 7 microexons.

(2) The phenotypic analysis using high-throughput approaches is sound and provides invaluable data.

We thank the reviewer for their support.

Weaknesses:

(1) There is not enough information on the exact nature of the deletion for each microexon.

To clarify the nature of our mutant alleles, we have added a figure (Figure S1) that details the location of the microexon in relation to its predicted neighboring exons, deletion boundaries,

guide RNAs, and putative regulatory elements.

(2) Only one deletion is phenotypically analysed, leaving space for the phenotype observed to be due to sequence modifications independent of the microexon itself.

We have determined whether the mRNA is impacted in unanticipated ways for a subset of mutants with mild phenotypes (see point #4 responses to Reviewer 1 for details). Our findings for three microexon mutants (ap1g1, vav2, and vti1a) are corroborated by LopezBlanch et al. (<https://doi.org/10.1101/2024.10.23.619860>). We have also already compared the microexon removal to a loss-of-function mutant for two lines (Figures S3 and S4), and we have made this comparison more obvious as well as increasing the discussion of the expected phenotypes from typical loss-of-function mutants (see point #3 response to reviewer 1).

Unlike protein-coding truncations, clean removal of the microexon and its regulatory elements is unlikely to yield different phenotypic outcomes if independent lines are generated (with the exception of genetic background effects). When generating a proteintruncating allele, the premature stop codon can have different locations and a varied impact on genetic compensation. In previous work (Capps et al., 2024), we have observed different amounts of nonsense-mediated decay-induced genetic compensation (El-Brolosy, et al., 2019) depending on the location of the mutation. As they lack variable premature stop codons (the expectation of a clean removal), two mutants for the same microexons should have equivalent impacts on the mRNA.

We now address the concern of subtle genetic background effects in the Methods: “Even with using sibling controls and collecting multiple biological replicates from individual parents, the possibility remains that linked genetic variation may have contributed to the mild phenotypes we observed, as only a single line was generated.”

Reviewer #3 (Public review):

Summary:

This paper sought to understand how microexons influence early brain function. By selectively deleting a large number of conserved microexons and then phenotyping the mutants with behavior and brain activity assays, the authors find that most microexons have minimal effects on the global brain activity and broad behaviors of the larval fish-- although a few do have phenotypes.

Strengths:

The work takes full advantage of the scale that is afforded in zebrafish, generating a large mutant collection that is missing microexons and systematically phenotyping them with high throughput behaviour and brain activity assays. The work lays an important foundation for future studies that seek to uncover the likely subtle roles that single microexons will play in shaping development and behavior.

We thank the reviewer for their support.

Weaknesses:

The work does not make it clear enough what deleting the microexon means, i.e. is it a clean removal of the microexon only, or are large pieces of the intron being removed as well-- and if so how much? Similarly, for the microexon deletions that do yield phenotypes, it will be important to demonstrate that the full-length transcript levels are unaffected by the deletion. For example, deleting the microexon might have unexpected

effects on splicing or expression levels of the rest of the transcript that are the actual cause of some of these phenotypes.

To clarify the nature of our mutant alleles, we have added a figure (Figure S1) that details the location of the microexon in relation to its predicted neighboring exons, deletion boundaries, guide RNAs, and putative regulatory elements. We have determined whether the mRNA is impacted in unanticipated ways for a subset of mutants with mild phenotypes (see point #4 responses to Reviewer 1 for details).

Reviewer #1 (Recommendations for the authors):

(1) For most ME mutations, 4 guide sequences are provided. More description / a diagram could be helpful to interpret how ME mutations were generated.

We have added diagrams to the Supplementary Materials (new Figure S1) to show where the guide RNAs, cut sites, and putative regulatory elements are in relationship to the microexon and its neighboring exons. We have also added the following point to the text: “Four guide RNAs were used, two on each side of the microexon (Table S2, Figure S1).”

(2) Figure 1 indicates that there are 45 microexons (MEs) but the text initially indicates that there are 44 that exist in a canonical layout (the text later indicates there are 45). This could be made more clear.

The 45 refers to the mutants that were generated, not the microexons with putative Srrm4 regulatory elements. We did not choose microexons to mutate based on whether they were regulated by Srrm4. We have clarified these points in the manuscript as follows: “Of these 95 microexons, 42 exist in a canonical layout in the zebrafish genome, with both a UGC and UC repeat – or similar polypyrimidine tract – directly upstream of the alternatively spliced exon (Gonatopoulos-Pournatzis et al., 2018) (Table S1), indicating that Srrm4 likely controls their inclusion. Of the remaining microexons, 44 are organized similarly to the canonical layout, typically with either a UGC or UC repeat. Thus, they may also be regulated by Srrm4.” and “Using CRISPR/Cas9, we generated lines that removed 45 conserved microexons (Table S2) and assayed larval brain activity, brain structure, and behavior (Figure 1A). Four guide RNAs were used, two on each side of the microexon (Table S2, Figure S1). For microexons with upstream regulatory elements that are likely important for splicing, these elements were also removed (Figure S1).”

(3) The description of the "canonical layout" as containing TGC / UC repeats could be rewritten as either "containing a UGC motif and UC repeats" or "containing a TGC motif and TC repeats."

This error has been corrected.

(4) Why was tcf7l2 selected as a control for MAP mapping?

The mutant for tcf7l2 is an example of a moderately strong phenotype from a recent study we completed (Capps et al., 2025). This mutant was selected because it has both increased and decreased activity and structure and is ideal for setting the range of the graph. We now include a comparison to additional mutants from this study of autism genes (Capps et al., 2025) to further demonstrate how mild the phenotypes are in the microexon removal mutants (new Figure 3G). We also include the activity and structure maps of tcf7l2 mutants in Supplementary Figures 9 and 11.

(5) What does it mean that of the remaining microexons, most are similar to canonical layout?

Typically, they would have one of the two regulatory elements instead of both or the location of the possible elements would be slightly farther away than expected. We have clarified this point in the manuscript as follows: “Of these 95 microexons, 42 exist in a canonical layout in the zebrafish genome, with both a UGC and UC repeat or similar polypyrimidine tract – directly upstream of the alternatively spliced exon (Gonatopoulos-Pournatzis et al., 2018) (Table S1), indicating that Srrm4 likely controls their inclusion. Of the remaining microexons, 44 are organized similarly to the canonical layout, typically with either a UGC or UC repeat. Thus, they may also be regulated by Srrm4.”

(6) Figure 2A is very difficult to see - most are either up or down - suggest splitting into 2 figures - one = heat map, second can summarize values that were both up and down.

We prefer to retain this information for accuracy. The bubble location is offset to effectively share the box between the orange (decreased) and purple (increased) measures. For example, and as noted in the methods and now expanded upon, a measure can change between 4 and 6 dpf or a measure such as bout velocity could be increased while the distance traveled is decreased (both are magnitude measures). The offset of the bubbles is consistently 0.2 data units in x and y from the center of the box.

(7) The authors apply rigorous approaches to testing the importance of microexons. I especially appreciate the inclusion of separate biological replicates in the main figures!

We thank the reviewer for their positive feedback.

(8) Page 5 line 5 - suggest "compared to homozygous mutants".

The change has been made.

(9) For *Eif5g3b* dark flash phenotype, it's not clear what "p-values are not calculated for response plots" means. A p-value is provided in the plot for *ppp6r3* response freq.

The *eif4g3b* plot is the actual response trace measuring through pixel changes whereas the *ppp6r3* is the frequency of response. While informative, the response plot is time-based data with a wide dynamic range, making the average signal across the entire time window meaningless. We include the p-values for a related measure, the latency for the first 10 dark flashes in block 1 (*day6dpfdf1a_responselatency*) in the legend.

(10) The *ptprd* phenotype in 2D is not described in the text.

The change has been made.

(11) Page 7 line 7: "mild" is repeated.

This error has been corrected.

Reviewer #2 (Recommendations for the authors):

Specific points for needed improvement:

(1) The title should be adjusted to more accurately describe the results. The term 'minimal' is under-representing the findings. 9/45 (20%) of targets in their screen have

some phenotype, indicating that a significant number have indeed an important function. Moreover, the phenotypic analysis is limited, leaving space for missed abnormalities (as discussed by the authors). I would therefore suggest a more neutral title such as 'Systematic genetic deletion of microexons uncovers their roles in zebrafish brain development and larval behaviour'.

While some microexon mutants do have repeatable phenotypes, these phenotypes are far milder than phenotypes observed in other mutant sets. We now include a comparison to additional mutants from this study of autism genes (Capps et al., 2025) to further demonstrate how mild the phenotypes are in the microexon removal mutants (new Figure 3G). The title states that these microexons have a minimal impact on larval zebrafish brain morphology and function, leaving room for the possibility of adult phenotypes. Thus, we prefer to retain this title.

(2) Do the 45 chosen microexons correspond to the 44 with a canonical layout with TGC and UC repeats? If so, it needs to be explicitly stated in the text that exons were chosen for mutation based on the potential for SRRM4 regulation. If not, then the rationale for the choice of the 45 mutants from the 95 highly conserved events needs to be explained further.

The 45 refers to the mutants that were generated, not the microexons with putative Srrm4 regulatory elements. We did not choose microexons to mutate based on whether they were regulated by Srrm4. We have clarified these points in the manuscript as follows: “Of these 95 microexons, 42 exist in a canonical layout in the zebrafish genome, with both a UGC and UC repeat – or similar polypyrimidine tract – directly upstream of the alternatively spliced exon (Gonatopoulos-Pournatzis et al., 2018) (Table S1), indicating that Srrm4 likely controls their inclusion. Of the remaining microexons, 44 are organized similarly to the canonical layout, typically with either a UGC or UC repeat. Thus, they may also be regulated by Srrm4.” and “Using CRISPR/Cas9, we generated lines that removed 45 conserved microexons (Table S2) and assayed larval brain activity, brain structure, and behavior (Figure 1A). Four guide RNAs were used, two on each side of the microexon (Table S2, Figure S1). For microexons with upstream regulatory elements that are likely important for splicing, these elements were also removed (Figure S1).”

There was no clear rationale for those that were selected. We attempted to generate all 95 and some mutants were not successfully generated in our initial attempt. As we found minimal phenotypes, we elected to not continue to make the remaining ones on the list.

(3) More detail regarding the design of guides for CRISPR is required in the text in the methods section. From Table S2, 4 guides were used per microexon. Were these designed to flank the microexon? How far into the intronic sequence were the guides designed? Were the splicing regulatory sequences (polypyrimidine tract, branchpoint) also removed? The flanking sequences of each of the 45 deletion lines need to be provided.

We have added diagrams to the Supplementary Materials (new Figure S1) to show where the guide RNAs, cut sites, and putative regulatory elements are in relationship to the microexon and its neighboring exons. We removed the microexon and the surrounding area that contains the putative regulatory elements.

(4) Following on from the previous point, to ascertain that the phenotype observed is truly due to lack of microexon (rather than other event linked to removed intronic sequences) - for the 7 exons newly identified as functionally important, at least one added deletion line has to be shown, presenting the same phenotype. If making 7 more lines can't be achieved in a reasonable time (we are aware this is a big ask), a MO

experiment blocking microexon splicing needs to be provided (may not be ideal for analysis at 6 dpf). For the existing mutants and the new ones (or morphants), sequencing of the mRNAs for the 7 genes in mutants and siblings also needs to be added to check any possible change in other variants.

Unlike protein-coding truncations, clean removal of the microexon and its regulatory elements is unlikely to yield different phenotypic outcomes if independent lines are generated (with the exception of genetic background effects). When generating a protein-truncating allele, the premature stop codon can have different locations and a varied impact on genetic compensation. In previous work (Capps et al., 2024), we have observed different amounts of nonsense-mediated decay-induced genetic compensation (El-Brolosy, et al., 2019) depending on the location of the mutation. As they lack variable premature stop codons (the expectation of a clean removal), two mutants for the same microexons should have equivalent impacts on the mRNA. We acknowledge that we inadequately described the generation of these alleles, and we now provide Figure S1 to show the microexon's relationship to possible regulatory elements that impact splicing in unexpected ways if they remain.

We now acknowledge the concern of subtle genetic background effects in the Methods: "Even with using sibling controls and collecting multiple biological replicates from individual parents, the possibility remains that linked genetic variation may have contributed to the mild phenotypes we observed, as only a single line was generated."

Given the caveats of MOs and transient microinjection for the study of 6 dpf phenotypes, we disagree that this suggested experiment would provide value. The phenotypic assays we use are highly sensitive, and we would not even trust CRISPRs to yield reliable data. We have added an additional loss-of-function allele for ppp6r3 from the Sanger knockout project, which has a similar but stronger size change to the ppp6r3 microexon-removal line. In addition, our findings for three microexon mutants (ap1g1, vav2, and vti1a) are corroborated by Lopez-Blanch et al. (<https://doi.org/10.1101/2024.10.23.619860>).

To support that these we generated clean removal of these microexons, we experimentally determined whether the mRNA is impacted in unanticipated ways for a subset of mutants with mild phenotypes (see the point #4 public response to Reviewer 1). We also have already compared the microexon removal to a loss-of-function mutant for two lines (Figure S1), and we have made that outcome more obvious as well as increasing the discussion of the expected phenotypes from typical loss-of-function mutants (see point #3 public response to Reviewer 1).

(5) Figure 3: An image of control tcf7l2 mutant brain activity as a reference should be included.

We now include the activity and structure maps of tcf7l2 mutants in Supplementary Figures 9 and 11.

(6) Figure 3a/b. The gene names on the y-axis of the pERK and structure comparisons should be reordered to be alphabetical so that phenotypes can be compared by the reader for the same microexon across the two assays.

These data are clustered so that any similarities between maps can be recognized. We prefer to retain the clustering to compare lines to each other.

(7) Figure S6 legend. Including graph titles like "day3msdf_dpix_numberofbouts_60" is not comprehensible to the reader so should be replaced with more descriptive text. As

should jargon such as "combo plot" and "habituation_day5dpfhab1post_responsefrequency_1_a1f1000d5p" etc.

The legend has been edited to describe the experiments. Subsections of the prior names are maintained in parentheses to enable the reader to connect the plots in this figure to the specific image and underlying data in Zenodo.

(8) Page 2 line 21 "to enable proper".

The change has been made.

(9) Page 7 line 7. Repeatable phenotypes were mild mild.

This error has been corrected.

Reviewer #3 (Recommendations for the authors):

(1) Figure 1B is confusingly laid out.

We are unclear how to modify Figure 1B, as it is a bar plot. We have modified several figures to improve clarity.

(2) Figure 1E-there are some pictures of zebrafish but to what end? They aren't labelled. The dark "no expression" looks really similar to the dark green, "high expression".

The zebrafish images represent the ages assessed for microexon inclusion. We have added labels to clarify this point.

(3) The main text says "microexons were removed by Crispr" but there is no detail in the main text about this at all-- and barely any in the methods. What does it mean to be removed? Cleanly? Or including part of the introns on either side? Etc. How selected, raised, etc? I can glean some of this from the Table S2 if I do a lot of extra work, but at least some notes about this would be important.

We have added diagrams to the Supplementary Materials (new Figure S1) to show where the guide RNAs, cut sites, and putative regulatory elements are in relationship to the microexon and its neighboring exons. We removed the microexon and the surrounding area that contains the putative regulatory elements.

(4) Figure 2 - There are no Ns, at least for the plots on the right. The reader shouldn't have to dig deep in Table S2 to find that. It is also unclear why heterozygous fish are not included in these analyses, since there are sibling data for all. Removed for readability of the plots might be warranted, but this should be made explicitly clear.

The Ns for these plots have been added to the legend. The legend was also modified as follows: "Comparisons to the heterozygous larvae are removed for clarity and available in the Supplementary Materials, as they often have even milder phenotypes than homozygous."

(5) Needed data: for those with phenotypes, some evidence should be presented that the full-length transcripts that encode proteins without the microexons are still expressed at the same level and without splicing errors/NMD. Otherwise, some of these phenotypes that were found could be due to knockdown or LOF (or I suppose even overexpression) of the targeted gene.

We have added a new Supplementary Figure S2 confirming clean removal of the microexons with RT-PCR for a subset of mutants with phenotypes. This figure also includes qRT-PCR for the same subset. We now discuss these findings: Results: “For eight mutant lines, we confirmed that the microexon was eliminated from the transcripts as expected (Figure S2). Although our genomic deletion did not yield unexpected isoforms, qRT-PCR on these eight lines revealed significant downregulation for the homozygous *vav2* mutant (Figure S2), indicating possibly complex genetic regulation.”

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